

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402875
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Bixby; Elliot et al.

Apparatus and method of tying tissue to bone

Abstract

A method of attaching target tissue to a pilot hole in a bone that utilizes a suture assembly having an anchor sleeve; a sliding suture, having a first end and a second end, and passing through the anchor sleeve; and a shuttle having a shuttle loop and a free end, opposed to the shuttle loop, and passing through the anchor sleeve, with the free end and the shuttle outside of the anchor sleeve. In the method, the sliding suture is passed about the target tissue and the first end is engaged to the shuttle loop. Then, the anchor sleeve is pushed into the pilot hole, and the free end is pulled to pull the shuttle into the anchor sleeve in the pilot hole, thereby setting the anchor sleeve into the pilot hole.

Inventors: Bixby; Elliot (Gladstone, OR), Anderson; Edwin (Ridgefield, WA), Cook; Nathan Daniel (Portland, OR), Ferguson; John Thomas (Portland, OR)

Applicant: RIVERPOINT MEDICAL, LLC (Portland, OR)

Family ID: 89123375

Assignee: Riverpoint Medical, LLC (Portland, OR)

Appl. No.: 18/377574

Filed: October 06, 2023

Related U.S. Application Data

continuation parent-doc US 17387854 20210728 US 11839367 child-doc US 18377574
us-provisional-application US 63057711 20200728

Publication Classification

Int. Cl.: A61B17/04 (20060101); A61B17/06 (20060101); A61F2/30 (20060101)

U.S. Cl.:

CPC **A61B17/0401** (20130101); **A61B17/0469** (20130101); **A61B17/0485** (20130101);
A61B17/06166 (20130101); **A61F2/30749** (20130101); A61B2017/0414 (20130101);
A61B2017/0445 (20130101); A61F2002/30751 (20130101)

Field of Classification Search

CPC: A61F (2/08); A61F (2/0811); A61F (2002/0852); A61B (17/0485); A61B (17/0401);
A61B (2017/0406); A61B (2017/0409); A61B (2017/0477); A61B (17/04); A61B
(17/06166); A61B (2017/0496); A61B (2017/06185); A61B (17/0469)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10849734	12/2019	Holowecky	N/A	A61B 17/0485
2001/0041916	12/2000	Bonutti	606/232	A61B 17/0401
2009/0326564	12/2008	White	606/139	A61B 17/06166
2014/0257382	12/2013	McCartney	606/232	A61B 17/0485
2017/0049432	12/2016	Dooney, Jr.	N/A	A61B 17/0487
2018/0221010	12/2017	Lund	N/A	A61B 17/0401
2018/0338756	12/2017	Black	N/A	A61B 17/0401
2021/0386418	12/2020	Dooney, Jr.	N/A	A61B 17/0401

Primary Examiner: Shi; Katherine M

Attorney, Agent or Firm: BrainSpark Associates, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/387,854 filed on Jul. 28, 2021, entitled “Apparatus and Method of Tying Tissue to Bone,” which claims the benefit of Provisional Appl. No. 63/057,711, filed Jul. 28, 2020, which are incorporated by reference in their entireties. This application also claims the benefit of Provisional Appl. No. 63/057,711, filed Jul. 28, 2020, which is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

(1) The labrum, the cup-shaped cartilage that lines and reinforces the glenoid (the bone defining the shallow shoulder socket), sometimes becomes torn or separates from the glenoid, thereby causing patient pain and limiting freedom of arm movement. To repair this tear or separation a number of methods are currently available to put a loop around a portion of the labrum, and to anchor that loop in the glenoid. It is typical to place three of these anchored loops along a portion of the labrum, to properly brace the labrum and/or reattach it to the glenoid.

(2) Initially, placing these anchored loops along a portion of the labrum required tying a knot for each loop, and leaving the knot in the patient, which tended to slow healing from the operation. More recently, knotless techniques have been developed. There is, however, still room for

improvement, for apparatuses and methods that permit the surgery to be performed more quickly and with a more certain and stable result.

SUMMARY

(3) The following embodiments and aspects thereof are described and illustrated in conjunction with systems, tools, and methods which are meant to be exemplary and illustrative, not limiting in scope. In various embodiments, one or more of the above-described problems have been reduced or eliminated, while other embodiments are directed to other improvements.

(4) In a first separate aspect, the present invention may take the form of a method of attaching target tissue to a pilot hole in a bone that uses a suture assembly having a distal and a proximal end, and including a suture length having a first portion, which terminates coincident to the distal end of the suture assembly, and a second portion, which is formed into a slip knot about the first portion, and an anchor sleeve, slidably engaged to the suture length, and co-located with the slip knot, the suture length being doubled upon itself, wherein first and second portions are adjacent and form a loop and a loop end where the suture length undergoes a U-bend, the loop end being coincident to the proximal end of the suture assembly. In the method, the adjacent first and second portions are passed about the target tissue and the loop end is passed about the distal end, so that a cinchable loop is formed from the loop end to where the first and second portions pass through the loop end. Then, the cinchable loop is cinched about the target tissue and the slip knot and anchor sleeve are slid, until they are adjacent to the target tissue. Finally, the slip knot and anchor sleeve are pushed into the pilot hole.

(5) In a second separate aspect, the present invention may take the form of a method of attaching target tissue to a pilot hole in a bone that utilizes a suture assembly having an anchor sleeve; a sliding suture, having a first end and a second end, and passing through the anchor sleeve; and a shuttle having a shuttle loop and a free end, opposed to the shuttle loop, and passing through the anchor sleeve, with the free end and the shuttle outside of the anchor sleeve. In the method, the sliding suture is passed about the target tissue and the first end is engaged to the shuttle loop. Then, the anchor sleeve is pushed into the pilot hole, and the free end is pulled to pull the shuttle into the anchor sleeve in the pilot hole, thereby setting the anchor sleeve into the pilot hole.

(6) In a third separate aspect, the present invention may take the form of a method of connecting two pieces of tissue that utilizes a repair assembly, which includes a repair piece, having a finger trap sleeve, a repair loop in fixed relationship to the sleeve and attached to a first end of the sleeve, and a repair strand extending out of a second side of the sleeve, opposed to the first end of the sleeve, the repair strand terminating in a repair strand free end; and a shuttle, including a shuttle loop and a shuttle strand that extends from the shuttle loop and passes through the sleeve, and terminating in a shuttle strand free end. In the method, the shuttle loop and the repair strand free end are passed through both of the two pieces of tissue, so that the repair loop and shuttle strand free end are separated from the shuttle loop and repair strand free end by the two pieces of tissue. Then, the repair strand free end is threaded through the repair loop and the shuttle loop, consecutively. The shuttle strand free end is pulled, thereby pulling the repair loop through the piece of tissue. Finally, the shuttle is pulled entirely free of the repair piece, and leaving the repair piece, looping through and about the two pieces of tissue, with the finger trap sleeve adjacent to the two pieces of tissue and the portion of the repair strand trapped in the finger trap sleeve.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Various embodiments of the invention are disclosed in the following detailed description and accompanying drawings.

(2) FIG. 1 is a side view of a drill creating a hole in bone, as a first step in a method for the

attachment of soft tissue to bone;

(3) FIG. 2 is a side view of a suture assembly being introduced through a cannulated instrument into the operation theater, as a second step in a method for the attachment of soft tissue to bone;

(4) FIG. 3 is a side view of the suture assembly being wrapped about soft tissue, as a third step in a method for the attachment of soft tissue to bone;

(5) FIG. 4 is a side view of the suture assembly being formed into a loop, as a fourth step in a method for the attachment of soft tissue to bone;

(6) FIG. 5 is a side view of the suture assembly, formed into a loop that is being cinched up around soft tissue, as a fifth step in a method for the attachment of soft tissue to bone;

(7) FIG. 6 is a side view of an insertion tool engaged with the suture assembly and aimed at the hold created in FIG. 1, as a sixth step in a method for the attachment of soft tissue to bone;

(8) FIG. 7 is a side view of the suture assembly, a portion of which has been pushed into the bone hole, being further cinched around the soft tissue, as a seventh step in a method for the attachment of soft tissue to bone;

(9) FIG. 8 is a side view of the suture assembly, now cinched tightly about the soft tissue, and with the portion pushed into the bone hole, with a portion now being pulled to securely anchor it in the bone hole;

(10) FIG. 9 is a side view of the suture assembly, with a portion cut off;

(11) FIG. 10A is a side view of the suture assembly of FIG. 2;

(12) FIG. 10B is a side view of a suture assembly, similar to that of FIG. 10A, but with a different slip knot and sleeve configuration;

(13) FIG. 10C is a side view of a suture assembly, similar to that of FIG. 10A, but with another different slip knot and sleeve configuration;

(14) FIG. 10D is a side view of a suture assembly, similar to that of FIG. 10A, but with yet another different slip knot and sleeve configuration;

(15) FIG. 10E is a side view of a suture assembly, similar to that of FIG. 10A, but with still another different slip knot and sleeve configuration;

(16) FIG. 11 is a side view of an alternative suture assembly;

(17) FIG. 12 is a side view of the alternative suture assembly of FIG. 11, in a first stage of its deployment attaching soft tissue to bone;

(18) FIG. 13 is a side view of the alternative suture assembly of FIG. 11, in a second stage of its deployment attaching soft tissue to bone;

(19) FIG. 14 is a side view of the alternative suture assembly of FIG. 11, in a third stage of its deployment attaching soft tissue to bone;

(20) FIG. 15 is a side view of the alternative suture assembly of FIG. 11, in the final stage of its deployment attaching soft tissue to bone;

(21) FIG. 16 is a side view of another alternative suture assembly;

(22) FIG. 17 is a side view of the alternative suture assembly of FIG. 16, in a first stage of its deployment attaching soft tissue to bone;

(23) FIG. 18 is a side view of the alternative suture assembly of FIG. 16, in a second stage of its deployment attaching soft tissue to bone;

(24) FIG. 19 is a side view of the alternative suture assembly of FIG. 16, in a third stage of its deployment attaching soft tissue to bone; and

(25) FIG. 20 is a side view of the alternative suture assembly of FIG. 16, in the final stage of its deployment attaching soft tissue to bone.

DETAILED DESCRIPTION AND EMBODIMENTS

(26) The following is a detailed description of exemplary embodiments to illustrate the principles of the invention. The embodiments are provided to illustrate aspects of the invention, but the invention is not limited to any embodiment. The scope of the invention encompasses numerous alternatives, modifications and equivalent; it is limited only by the claims.

(27) Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. However, the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

(28) In a first preferred embodiment, target tissue **12** is tied to a nearby bone **14**, utilizing in part a cannula tool **16**, which is first used to introduce a drill **18** to create a pilot hole **20** in bone **14**. Referring, now, to FIG. 2 a suture assembly **22**, having a distal end **24** and a proximal end **26**, includes a suture length **28**, having a first portion **29** that terminates in a location coincident to the distal end **24** of the assembly **22**. Suture length **28** is doubled up upon itself, thereby forming a loop end that is coincident with the proximal end **26** of assembly **22**. A second portion **30** of suture length is formed into a slip knot **32** about the first portion **29**. Slip knot **32** is positioned within the lumen of a braided suture **36**, that forms an anchor sleeve.

(29) Referring to FIG. 3, proximal end **26** is brought around a portion of target tissue **12** and, referring to FIG. 4, looped over the distal end **24** of the suture assembly **22**, thereby forming a cinchable loop **40**, that extends from proximal end **26**, around target tissue **12**, and up to the point **42** where first portion **29** extends past proximal end **26**. Next, referring to FIGS. 5 and 6, the loop **40** is cinched up by pulling on the distal end **24** of assembly **22**, and introducer **44** is used to push anchor sleeve **36**, and included slip knot **32**, into pilot hole **20**. As shown in FIGS. 7-9, introducer tool **44** is removed, first portion **29** is pulled, setting anchor sleeve **36** into pilot hole **20**, and the remainder of first portion **29** is trimmed off, leaving a knotless binding of target tissue **12** to bone **14**.

(30) Referring, now, to FIGS. 10A-10E, a number of variants of assembly **22** are shown, having differences in the anchor sleeve **36** and slip knot **32** arrangement. FIG. 10B shows assembly **221**, in which, going from distal end to proximal end, first portion **29** enters the distal end of anchor sleeve **361**, broaches out of anchor sleeve **361**, where it is joined by second portion **30**, tied in slip knot **32**, and then broaches into anchor sleeve **361** before exiting the proximal end of the lumen of anchor sleeve **361**. FIG. 10C shows assembly **222**, in which, again from distal end to proximal end, first portion **29** enters anchor sleeve **362**, is joined by second portion **30** tied in a slip knot **32** about portion **29**, and then broaches back and forth out of and into anchor sleeve **362**, forming two undulations **365** out of anchor sleeve **362**, before reentering and exiting out of the lumen proximal end. The undulations **365** help to set anchor firmly into pilot hole **20**. Other numbers of undulations **365** are possible. In FIGS. 10D and 10E assemblies **223** and **224** are the same as assembly **22**, except for that in assembly **223** slip knot **32** is held in place with a spot of glue **366** and in assembly **224**, slip knot **32** is held in place with a stitch **367**.

(31) Referring to FIGS. 11-15, in a second preferred embodiment an anchored-loop forming assembly **110** is provided, including an anchor sleeve **112**, and a sliding suture **114** having a first end **116** and a second end **118**. Also, a shuttle **117** is engaged through sleeve **112** and includes a shuttle loop **119**. Referring to FIG. 12, the method begins with the positioning of suture **114** about a portion of target tissue **120**, and the engagement of the first end **116** to the shuttle loop **119** introduction of sleeve **112** into a pilot hole **122**. (At this point, the pilot hole **122** has already been drilled). The anchor sleeve **112** is pushed into the pilot hole **122** (using a tool similar to introducer tool **44**, shown in FIG. 6). As shown in FIG. 14, the free end of shuttle **117** is pulled to pull shuttle loop **119** into anchor sleeve, where it acts to thicken anchor sleeve **112**, setting it into pilot hole **122** and the stray ends of suture **114** and shuttle **117** are cut (as shown in FIG. 15). The second embodiment utilizes the same sort of cannulated instrument **16** as shown in the figures describing the first embodiment. In one embodiment the shuttle loop **119** is made of a metal, such as nitinol that is roughened so that said implantable tail physically engages to said shuttle loop **119**. In this embodiment the shuttle **117** may be made of metal, such as nitinol.

(32) In both the first and second embodiments, target tissue **12**, **120** is typically pierced with a

needle, to pull the suture about it. One application of these described methods/embodiments is the reattachment of the labrum to the glenoid, in the shoulder of a patient.

(33) Referring to FIGS. **16-20**, in a third embodiment and method, a construct **410**, includes an implantable part **412**, including a loop, **414**, connected to a sleeve **416**, which is connected to implantable tail **418**. Engaged with implantable part **412** is a shuttle **420**, including a shuttle loop **422**, attached to a shuttle tail **424**. Tails **418** and **424** are made of suture material. Referring to FIG. **17**, construct **410** has been engaged to two sections of tissue **450** and **452**, using a suture passer **454**. Suture passer **454** may be, for example, the NanoPass, available from Stryker Corporation, which maintains a website at www.stryker.com. Referring to FIG. **18**, implantable tail **418** is engaged through implantable loop **414** and shuttle loop **422**. The shuttle tail **424** is pulled, thereby as shown in FIG. **19**, pulling implantable loop **414** and sleeve **416** through tissue **452**; pulling implantable tail **418** through sleeve **416** and tissue **450** and pulling shuttle **420** completely out of the construct **410**. Implantable loop **414** is thereby formed into a loop holding tissues **450** and **452**, together. Tail **418** is then pulled further, and as shown in FIG. **20**, implantable part **412** is cinched tight, to better hold tissues **450** and **452** together. In one embodiment the shuttle loop **422** is made of a metal, such as nitinol that is roughened so that said implantable tail physically engages to said shuttle loop **422**. In this embodiment the shuttle **420** may be made of metal, such as nitinol.

(34) In an alternative embodiment, tissue (such as tissue **450**) is attached to bone, by pushing sleeve **416** into a pilot hole in bone, similar to pushing sleeve **112** into pilot hole **122** in FIG. **14**, or sleeve **36** into pilot hole **20**, in FIG. **9**. Then, tail **418** can be pulled to tighten implantable part **412**, to hold tissue **450** to the bone.

(35) The above-described constructs and methods find application, for example, in repairing a tear in the meniscus of the knee joint or reattaching a portion of the labrum to the glenoid in the shoulder joint.

(36) The disclosed embodiments are illustrative, not restrictive. While specific configurations of the suture assemblies and their methods of use have been described, it is understood that the present invention can be applied to a wide variety of apparatuses and methods of repair of tendons and other body tissue. There are many alternative ways of implementing the invention.

Claims

1. A method of attaching target tissue to a pilot hole in a bone, comprising: providing a suture assembly, comprising: i. a flexible suture anchor sleeve, ii. a sliding suture, having a first end and a second end, and engaging with said flexible suture anchor sleeve; and iii. a shuttle having a shuttle loop and a free end, said free end opposed to said shuttle loop, a portion of said shuttle engaging with said flexible suture anchor sleeve, said free end and said shuttle loop extending outside of said flexible suture anchor sleeve; pushing said flexible suture anchor sleeve into said pilot hole; passing a first end of said sliding suture around said target tissue and engaging said first end to said shuttle loop to create a cinchable loop; pulling on said free end of said shuttle to pull said shuttle loop and a portion of said first end of sliding suture into said flexible suture anchor sleeve; and retaining a least a portion of said shuttle loop within said flexible suture anchor sleeve to wedge said portion of said shuttle loop and a portion of said first end of said sliding suture within said flexible suture anchor sleeve, wherein said suture assembly holds said target tissue to bone.
2. The method of claim 1, wherein said shuttle loop comprises nitinol.
3. The method of claim 1, wherein said bone is the glenoid and said target tissue is a portion of the labrum.
4. The method of claim 1, wherein said method further includes a step that comprises setting said flexible suture anchor sleeve into said pilot hole.
5. The method of claim 4, wherein said setting comprises thickening of said flexible suture anchor sleeve within said pilot hole.

6. The method of claim 1, wherein a portion of said shuttle loop comprises a roughened surface.
 7. The method of claim 1, wherein said bone comprises a portion of a knee and said target tissue comprises a meniscus.
 8. The method of claim 1, wherein a portion of said shuttle comprises a suture material.
 9. The method of claim 1, wherein the step of pulling on said free end of said shuttle through said flexible suture anchor sleeve comprises closing said cinchable loop to contact said target tissue.
 10. The method of claim 1, wherein said flexible suture anchor sleeve comprises a braided suture.
 11. The method of claim 1, wherein said method further includes cutting said second end of said sliding suture and free end of said shuttle.
-